

Topics to discuss

- Derivation of Newton's Cotes formula
- Derivation of Trapezoidal Rule
- Derivation of Simpson's $\frac{1}{3}$ Rule
- Derivation of Simpson's $\frac{3}{8}$ Rule
- Derivation of Weddle's Rule

Newton-Cotes Methods :

The derivation of Newton's cotes formula is based on polynomial interpolation

an n^{th} degree polynomial

$p_n(x)$ that interpolates the values of $f(x)$ at $n+1$

evenly spaced points can be used to replace

the integrand $f(x)$ of the integral

$$I = \int_a^b f(x) dx$$

and the resultant formula is called $(n+1)$ point Newton-Cotes formula.

Derivation: Let $y = f(x)$ be a function such that, $y_0 = f(x_0)$, $y_1 = f(x_1)$, $y_2 = f(x_2)$, ..., $y_n = f(x_n)$

where, $x_n = x_0 + nh$

By Newton's forward difference interpolation formula,

$$y = f(x) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0 + \dots$$

where,

$$u = \frac{x - x_0}{h} \Rightarrow x = x_0 + uh$$

Now let's integrate, $y = f(x)$

$$\int_{x_0}^{x_n} y dx = \int_{x_0}^{x_0 + nh} f(x) dx \quad \text{--- (i)}$$

We know,

$$x = x_0 + uh$$

$$dx = h du$$

also, if $x = x_0 \Rightarrow u = 0$

and if $x = x_0 + nh \Rightarrow u = n$

Now eqn-① becomes,

$$\int_{x_0}^{x_n} y dx = \int_{x_0}^{x_0+nh} f(x) dx \quad \text{--- ①}$$

$$\Rightarrow \int_{x_0}^{x_n} y dx = h \int_0^n f(x_0 + uh) du$$

$$= h \int_0^n \left[y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0 + \dots \right] du$$

$$\begin{aligned}
\Rightarrow \int_{x_0}^{x_n} y dx &= h \int_0^n \left[y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0 + \dots \right] du \\
&= h \int_0^n \left[y_0 + u \Delta y_0 + \frac{1}{2} (u^2 - u) \Delta^2 y_0 + \frac{1}{6} (u^3 - 3u^2 + 2u) \Delta^3 y_0 + \dots \right] du \\
&= h \left[u y_0 + \frac{u^2}{2} \Delta y_0 + \frac{1}{2} \left(\frac{u^3}{3} - \frac{u^2}{2} \right) \Delta^2 y_0 + \frac{1}{6} \left(\frac{u^4}{4} - \frac{3u^3}{3} + \frac{2u^2}{2} \right) \Delta^3 y_0 + \dots \right]_0^n \\
&= h \left[n y_0 + \frac{n^2}{2} \Delta y_0 + \frac{1}{12} (2n^3 - 3n^2) \Delta^2 y_0 + \frac{1}{24} (n^4 - 4n^3 + 4n^2) \Delta^3 y_0 + \dots \right] \\
&= nh \left[y_0 + \frac{n}{2} \Delta y_0 + \frac{1}{12} (2n^2 - 3n) \Delta^2 y_0 + \frac{1}{24} (n^3 - 4n^2 + 4n) \Delta^3 y_0 + \dots \right] \\
&= nh \left[y_0 + \frac{n}{2} \Delta y_0 + \frac{n(2n-3)}{12} \Delta^2 y_0 + \frac{n(n-2)^2}{24} \Delta^3 y_0 + \dots \right]
\end{aligned}$$

This is called general Newton's Cotes's formula.

If the limits of integration a and b are in the set of interpolating points $x_i, i=0, 1, 2 \dots n$ then the formula is referred as closed form.

If the points a and b lie beyond the set of interpolating points, then the formula is termed open form. Since open form formula is not used for definite integration, we consider here only the closed form methods. They include

1. Trapezoidal Rule (two-point formula), $n=1$
2. Simpson's $1/3$ Rule (three-point formula), $n=2$
3. Simpson's $3/8$ Rule (four-point formula), $n=3$
4. Boole's Rule (five-point formula), $n=4$
5. Weddle's Rule (seven-point formula), $n=6$

Derivation of Trapezoidal Rule :

For $n=1$, means we have two points x_0 & x_1

So, table will be like

x	y	Δy
x_0	y_0	Δy_0
x_1	y_1	

Now, for $n=1$, Newton Cote's formula will become,

$$\int_{x_0}^{x_1} y dx = nh \left[y_0 + \frac{n}{2} \Delta y_0 \right]$$
$$= h \left[y_0 + \frac{1}{2} (y_1 - y_0) \right]$$

$$\int_{x_0}^{x_1} y dx = \frac{h}{2} [y_0 + y_1]$$

Similarly,

$$\int_{x_1}^{x_2} y dx = \frac{h}{2} [y_1 + y_2]$$

$$\int_{x_2}^{x_3} y dx = \frac{h}{2} [y_2 + y_3]$$

⋮

$$\int_{x_{n-1}}^{x_n} y dx = \frac{h}{2} [y_{n-1} + y_n]$$

So, we can write,

$$\begin{aligned} \int_{x_0}^{x_n} y dx &= \int_{x_0}^{x_1} y dx + \int_{x_1}^{x_2} y dx + \int_{x_2}^{x_3} y dx + \dots + \int_{x_{n-1}}^{x_n} y dx \\ &= \frac{h}{2} [y_0 + y_1 + y_1 + y_2 + y_2 + y_3 + \dots + y_{n-1} + y_n] \\ &= \frac{h}{2} [y_0 + 2(y_1 + y_2 + y_3 + \dots + y_{n-1}) + y_n] \end{aligned}$$

This is known as Trapezoidal Rule.

Derivation of Simpson's 1/3 Rule :

for $n=2 \Rightarrow$ we have three points, x_0, x_1, x_2
so, forward difference table,

x	y	Δy	$\Delta^2 y$
x_0	y_0	Δy_0	$\Delta^2 y_0$
x_1	y_1	Δy_1	
x_2	y_2		

Now taking case of $n=2$ in Newton's Cotes formula, we get,

$$\begin{aligned}\int_{x_0}^{x_2} y dx &= nh \left[y_0 + \frac{n}{2} \Delta y_0 + \frac{n(2n-3)}{12} \Delta^2 y_0 \right] \\ &= 2h \left[y_0 + \Delta y_0 + \frac{1}{6} \Delta(\Delta y_0) \right] \\ &= 2h \left[y_0 + y_1 - y_0 + \frac{1}{6} \Delta(y_1 - y_0) \right] \\ &= 2h \left[y_1 + \frac{1}{6} (y_2 - y_1 - y_1 + y_0) \right] \\ &= \frac{2h}{6} [6y_1 + y_2 - 2y_1 + y_0] \\ &= \frac{h}{3} [y_0 + 4y_1 + y_2]\end{aligned}$$

Similarly,

$$\int_{x_2}^{x_4} y dx = \frac{h}{3} [y_2 + 4y_3 + y_4]$$

$$\int_{x_4}^{x_6} y dx = \frac{h}{3} [y_4 + 4y_5 + y_6]$$

$$\vdots$$
$$\int_{x_{n-2}}^{x_n} y dx = \frac{h}{3} [y_{n-2} + 4y_{n-1} + y_n]$$

So, we can write,

$$\int_{x_0}^{x_n} y dx = \int_{x_0}^{x_2} y dx + \int_{x_2}^{x_4} y dx + \int_{x_4}^{x_6} y dx + \dots + \int_{x_{n-2}}^{x_n} y dx.$$

$$= \frac{h}{3} [y_0 + 4y_1 + y_2 + y_2 + 4y_3 + y_4 + y_4 + 4y_5 + y_6 + \dots + y_{n-2} + 4y_{n-1} + y_n]$$
$$= \frac{h}{3} [y_0 + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \dots + y_{n-2}) + y_n]$$

This is known as Simpson's $\frac{1}{3}$ Rule.

Derivation of Simpson's 3/8 Rule

if $n=3$ then, Newton's cotes formula become,

$$\int_{x_0}^{x_3} y dx = nh \left[y_0 + \frac{n}{2} \Delta y_0 + \frac{n(n-3)}{12} \Delta^2 y_0 + \frac{n(n-2)^2}{24} \Delta^3 y_0 \right]$$

$$= 3h \left[y_0 + \frac{3}{2} \Delta y_0 + \frac{3}{4} \Delta(y_1 - y_0) + \frac{1}{8} \Delta(\Delta(y_1 - y_0)) \right]$$

$$= 3h \left(y_0 + \frac{3}{2} \Delta y_0 + \frac{3}{4} (y_2 - y_1 - y_1 + y_0) + \frac{1}{8} \Delta(y_2 - y_1 - y_1 + y_0) \right)$$

$$= 3h \left(y_0 + \frac{3}{2} \Delta y_0 + \frac{3}{4} (y_2 - 2y_1 + y_0) + \frac{1}{8} (y_3 - y_2 - y_2 + y_1 - y_2 + y_1 + y_1 - y_0) \right)$$

$$= 3h \left(y_0 + \frac{3}{2} \Delta y_0 + \frac{3}{4} (y_2 - 2y_1 + y_0) + \frac{1}{8} (y_3 - 3y_2 + 3y_1 - y_0) \right)$$

$$\begin{aligned}
&= 3h \left(y_0 + \frac{3}{2} \Delta y_0 + \frac{3}{4} (y_2 - 2y_1 + y_0) + \frac{1}{8} (y_3 - 3y_2 + 3y_1 - y_0) \right) \\
&= \frac{3h}{8} (8y_0 + 12\Delta y_0 + 6y_2 - 12y_1 + 6y_0 + y_3 - 3y_2 + 3y_1 - y_0) \\
&= \frac{3h}{8} (13y_0 + 12y_1 - 12y_0 + 6y_2 - 12y_1 + y_3 - 3y_2 + 3y_1) \\
&= \frac{3h}{8} (y_0 + 3y_1 + 3y_2 + y_3) \\
&= \frac{3h}{8} (y_0 + 3(y_1 + y_2) + y_3)
\end{aligned}$$

Similarly, $\int_{x_3}^{x_6} y \, dx = \frac{3h}{8} (y_3 + 3(y_4 + y_5) + y_6)$

and so on... $\int_{x_{n-3}}^{x_n} y \, dx = \frac{3h}{8} (y_{n-3} + 3(y_{n-2} + y_{n-1}) + y_n)$

Therefore,

$$\int_{x_0}^{x_n} y dx = \int_{x_0}^{x_3} y dx + \int_{x_3}^{x_6} y dx + \dots + \int_{x_{n-3}}^{x_n} y dx$$

$$= \frac{3h}{8} \left[(y_0 + 3y_1 + 3y_2 + y_3) + (y_3 + 3y_4 + 3y_5 + y_6) \right. \\ \left. + \dots + (y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n) \right]$$

$$= \frac{3h}{8} \left(y_0 + 3(y_1 + y_2 + y_4 + y_5 + \dots + y_{n-2} + y_{n-1}) + \right. \\ \left. 2(y_3 + y_6 + \dots + y_{n-3}) + y_n \right)$$

This is known as Simpson's $\frac{3}{8}$ rule.

Boole's Rule :

If we take the case of $n=4$
then we will get Boole's Rule

$$\int_{x_0}^{x_n} y dx = \frac{2h}{45} (7y_0 + 32y_1 + 12y_2 + 32y_3 + 14y_4 + 32y_5 + 12y_6 + 32y_7 + 14y_8 + \dots)$$

where, n is the multiple of 4.

This is known as boole's rule.

Weddle's Rule

If we take case of $n=6$ in Newton's cot's formula, we will get Weddle's rule,

$$\int_{x_0}^{x_n} y dx = \frac{3h}{10} \left[(y_0 + 5y_1) + (y_2 + 6y_3) + (y_4 + 5y_5) + y_6 + \right. \\ (y_6 + 5y_7) + (y_8 + 6y_9) + (y_{10} + 5y_{11}) + y_{12} + \\ (y_{12} + 5y_{13}) + (y_{14} + 6y_{15}) + (y_{16} + 5y_{17}) + y_{18} \\ \left. + \dots \right]$$

* n must be multiple of 6.

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